

Intro to ML/ Model Building

Week- 4

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What we have done so far ????

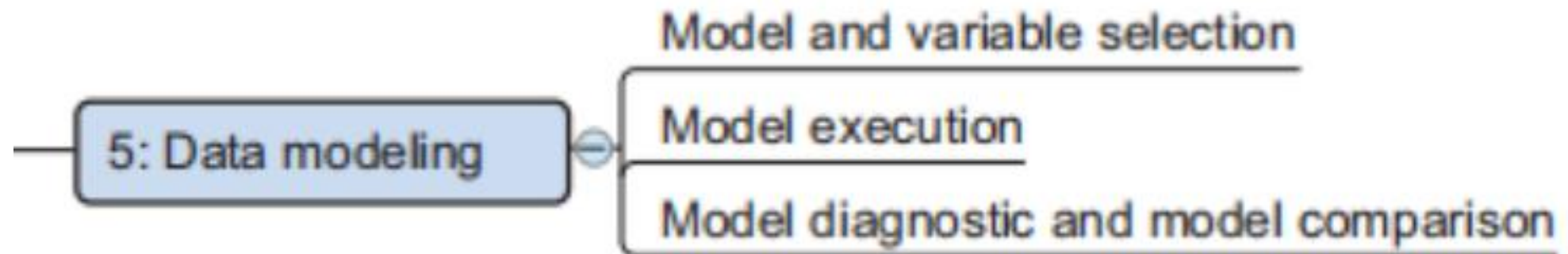
Problem definition → Data collection → Data cleaning → EDA

NOW WHAT???

“Now that we understand the data, the next step is to let the computer **learn patterns** from it — this is Model Building.”

What is a Model???

- A **model** is just a **mathematical representation of reality**.
- **Model building** = using algorithms to find patterns in data so we can make predictions or decisions.
- Input : Data
- Algorithm : Learn Patterns
- Output: Predictions/Decisions



Data Modeling Steps

The modelling phase consists of four steps:

1. Feature engineering and model selection
2. Training the model
3. Model validation and selection
4. Applying the trained model to unseen data

- It's possible to *chain* or *combine* multiple techniques. When you chain multiple models, the output of the first model becomes an input for the second model. When you combine multiple models, you train them independently and combine their results.
- A model consists of constructs of information called *features* or *predictors* and a *target* or *response variable*

Types of Models

- **Regression models** → predict numbers (e.g., predicting house price).
- **Classification models** → predict categories (e.g., spam vs. not spam).
- **Clustering models** → group similar items (e.g., customer segmentation).

Regression Model

- finding a relationship between variables. Regression is about **predicting a number**.
- Example:
- *“If you study more, you usually score higher.”*
- *“If you spend more money on ads, sales increase.”*

A way to predict a continuous value based on input data.

Examples:

- Predicting marks from study hours
- Predicting house price from size
- Predicting temperature from altitude

PACKAGES FOR WORKING WITH DATA IN MEMORY

When prototyping, the following packages can get you started by providing advanced functionalities with a few lines of code:

- *SciPy* is a library that integrates fundamental packages often used in scientific computing such as NumPy, matplotlib, Pandas, and SymPy.
- *NumPy* gives you access to powerful array functions and linear algebra functions.
- *Matplotlib* is a popular 2D plotting package with some 3D functionality.
- *Pandas* is a high-performance, but easy-to-use, data-wrangling package. It introduces dataframes to Python, a type of in-memory data table. It's a concept that should sound familiar to regular users of R.
- *SymPy* is a package used for symbolic mathematics and computer algebra.
- *StatsModels* is a package for statistical methods and algorithms.
- *Scikit-learn* is a library filled with machine learning algorithms.
- *RPy2* allows you to call R functions from within Python. R is a popular open source statistics program.
- *NLTK* (Natural Language Toolkit) is a Python toolkit with a focus on text analytics.

Linear Regression-Mathematical Model

- Linear regression is about finding a **line** that best describes the relationship between two variables.

$$Y = \beta_0 + \beta_1 X + \epsilon$$

- Y = dependent variable (marks)
- X = independent variable (hours studied)
- β_0 = intercept (marks when hours = 0)
- β_1 = slope (increase in marks per extra hour studied)
- ϵ = error term (unexplained variation)

Example- Solve yourself

Hours studied (X)	Marks (Y)
1	35
2	40
3	50
4	55
5	65

$$\beta_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x}$$

- Mean of X: $\bar{x} = (1 + 2 + 3 + 4 + 5)/5 = 3$
- Mean of Y: $\bar{y} = (35 + 40 + 50 + 55 + 65)/5 = 49$

$$\begin{aligned}\beta_1 &= \frac{(1 - 3)(35 - 49) + (2 - 3)(40 - 49) + (3 - 3)(50 - 49) + (4 - 3)(55 - 49) + (5 - 3)(65 - 49)}{(1 - 3)^2 + (2 - 3)^2 + (3 - 3)^2 + (4 - 3)^2 + (5 - 3)^2} \\ &= \frac{(-2)(-14) + (-1)(-9) + (0)(1) + (1)(6) + (2)(16)}{4 + 1 + 0 + 1 + 4} \\ &= \frac{28 + 9 + 0 + 6 + 32}{10} = \frac{75}{10} = 7.5\end{aligned}$$

Now intercept:

$$\beta_0 = 49 - 7.5 \times 3 = 49 - 22.5 = 26.5$$

$$Y = 26.5 + 7.5X$$

Interpretation:

- **Intercept (26.5):** If a student studies 0 hours, expected marks ≈ 26.5 .
- **Slope (7.5):** Each extra hour of study increases marks by ≈ 7.5 .

If a student studies 6 hours:

$$Y = 26.5 + 7.5(6) = 26.5 + 45 = 71.5$$

So, predicted marks ≈ 72 .

Multiple Linear Regression

- More than one input
- Multiple Linear Regression (MLR) is a way to **predict one thing (the outcome)** using **several factors (the inputs)**.

$$Y = b_0 + b_1X_1 + b_2X_2 + b_3X_3 + \dots + b_nX_n$$

Y → the thing you want to predict (e.g., house price).

$X_1, X_2, X_3, \dots$ → the factors that affect it (e.g., size, location, number of bedrooms).

b_0 → the starting value (intercept).

$b_1, b_2, b_3, \dots$ → weights (coefficients) that tell how much each factor matters.

MLR-Example

- Suppose you want to predict **student marks**. The outcome depends on:
- **Hours studied (X_1)**
- **Sleep hours (X_2)**
- **Past GPA (X_3)**

$$\text{Marks} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

Student	Hours studied (X_1)	Sleep hours (X_2)	GPA (X_3)	Marks (Y)
A	2	6	2.5	60
B	4	5	3.0	75
C	5	7	3.2	85
D	3	8	2.8	70
E	6	6	3.5	90

$$\text{Marks} = b_0 + b_1X_1 + b_2X_2 + b_3X_3$$

b_0 = intercept (base marks if all factors are zero)

b_1 = effect of **hours studied**

b_2 = effect of **sleep hours**

b_3 = effect of **GPA**

The computer will try to fit the regression equation where b values are learned from the data. We want to find the **best values of $b_0, b_1, b_2, \dots$** so that predictions are as close as possible to the real marks.

We use **Least Squares Method** → minimize the **errors** (differences between actual marks and predicted marks).

$$\text{Error} = Y_{\text{actual}} - Y_{\text{predicted}}$$

We try to make the **sum of squared errors (SSE)** as small as possible:

$$SSE = \sum (Y - \hat{Y})^2$$

If we write the data in **matrix form**:

- Y = vector of outcomes (marks)
- X = matrix of inputs (with a column of 1s for b_0)
- B = vector of coefficients ($b_0, b_1, b_2, \dots$)

Then the equation is:

$$\hat{Y} = XB$$

The solution for B is:

$$B = (X^T X)^{-1} X^T Y$$

This formula comes from calculus (minimizing SSE).

Say we had **only Hours studied** (X_1) $\rightarrow$ simple regression:

$$\text{Marks} = b_0 + b_1 X_1$$

Then:

$$b_1 = \frac{\sum (X_i - \bar{X})(Y_i - \bar{Y})}{\sum (X_i - \bar{X})^2}$$

$$b_0 = \bar{Y} - b_1 \bar{X}$$

$$B = (X^T X)^{-1} X^T Y$$

$$B = (b_0, b_1, b_2, b_3)^\top = (X^\top X)^{-1} X^\top y$$

We have 5 students; columns are: (Intercept), Hours, Sleep, GPA.

$$X_{\text{design}} = \begin{bmatrix} 1 & 2 & 6 & 2.5 \\ 1 & 4 & 5 & 3.0 \\ 1 & 5 & 7 & 3.2 \\ 1 & 3 & 8 & 2.8 \\ 1 & 6 & 6 & 3.5 \end{bmatrix} \quad y = \begin{bmatrix} 60 \\ 75 \\ 85 \\ 70 \\ 90 \end{bmatrix}$$

(Checks: sums — $n = 5$, $\sum \text{Hours} = 20$, $\sum \text{Sleep} = 32$, $\sum \text{GPA} = 15$, $\sum y = 380$.)

Compute elementwise (every element is a sum over rows):

- $(1, 1) = \sum 1 = 5$
- $(1, 2) = \sum \text{Hours} = 20$
- $(1, 3) = \sum \text{Sleep} = 32$
- $(1, 4) = \sum \text{GPA} = 15$
- $(2, 2) = \sum \text{Hours}^2 = 90$
- $(2, 3) = \sum \text{Hours} \cdot \text{Sleep} = 127$
- $(2, 4) = \sum \text{Hours} \cdot \text{GPA} = 62.4$
- $(3, 3) = \sum \text{Sleep}^2 = 210$
- $(3, 4) = \sum \text{Sleep} \cdot \text{GPA} = 95.8$
- $(4, 4) = \sum \text{GPA}^2 = 45.58$

$$X^T X = \begin{bmatrix} 5 & 20 & 32 & 15 \\ 20 & 90 & 127 & 62.4 \\ 32 & 127 & 210 & 95.8 \\ 15 & 62.4 & 95.8 & 45.58 \end{bmatrix}$$

Each entry is a sum-product:

- $\sum y = 380$
- $\sum \text{Hours} \cdot y = 1595$
- $\sum \text{Sleep} \cdot y = 2430$
- $\sum \text{GPA} \cdot y = 1158$

$$X^T y = \begin{bmatrix} 380 \\ 1595 \\ 2430 \\ 1158 \end{bmatrix}$$

For a 2x2 matrix the inverse is:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}^{-1} = \frac{1}{4 \times 6 - 7 \times 2} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \frac{1}{10} \begin{bmatrix} 6 & -7 \\ -2 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix}$$

$$\begin{aligned} \begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix} \begin{bmatrix} 0.6 & -0.7 \\ -0.2 & 0.4 \end{bmatrix} &= \begin{bmatrix} 4 \times 0.6 + 7 \times -0.2 & 4 \times -0.7 + 7 \times 0.4 \\ 2 \times 0.6 + 6 \times -0.2 & 2 \times -0.7 + 6 \times 0.4 \end{bmatrix} \\ &= \begin{bmatrix} 2.4 - 1.4 & -2.8 + 2.8 \\ 1.2 - 1.2 & -1.4 + 2.4 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

$$(X^{\top}X)^{-1} \approx \begin{bmatrix} 1081.553188 & 129.234043 & 2.893617 & -538.936170 \\ 129.234043 & 15.829787 & 0.531915 & -65.319149 \\ 2.893617 & 0.531915 & 0.212766 & -2.127660 \\ -538.936170 & -65.319149 & -2.127660 & 271.276596 \end{bmatrix}$$

Compute each coefficient β_j as the dot product of row j of $(X^{\top}X)^{-1}$ with $X^{\top}y$.

$$(X^{\top}X)^{-1} \approx \begin{bmatrix} 1081.553188 & 129.234043 & 2.893617 & -538.936170 \\ 129.234043 & 15.829787 & 0.531915 & -65.319149 \\ 2.893617 & 0.531915 & 0.212766 & -2.127660 \\ -538.936170 & -65.319149 & -2.127660 & 271.276596 \end{bmatrix} \quad X^{\top}y = \begin{bmatrix} 380 \\ 1595 \\ 2430 \\ 1158 \end{bmatrix}$$

b_0 (intercept)

$$\begin{aligned} b_0 &= 1081.553188 \cdot 380 + 129.234043 \cdot 1595 + 2.893617 \cdot 2430 + (-538.936170) \cdot 1158 \\ &= 410990.212766 + 206128.297872 + 7031.489362 - 624088.085107 \\ &\approx 61.914894 \end{aligned}$$

b_1 (Hours studied)

$$\begin{aligned} b_1 &= 129.234043 \cdot 380 + 15.829787 \cdot 1595 + 0.531915 \cdot 2430 + (-65.319149) \cdot 1158 \\ &= 49108.936170 + 25248.510638 + 1292.553191 - 75639.574468 \\ &\approx 10.425532 \end{aligned}$$

b_2 (Sleep hours)

$$\begin{aligned} b_2 &= 2.893617 \cdot 380 + 0.531915 \cdot 1595 + 0.212766 \cdot 2430 + (-2.127660) \cdot 1158 \\ &= 1099.574468 + 848.404255 + 517.021277 - 2463.829787 \\ &\approx 1.170213 \end{aligned}$$

b_3 (GPA)

$$\begin{aligned} b_3 &= -538.936170 \cdot 380 + (-65.319149) \cdot 1595 + (-2.127660) \cdot 2430 + 271.276596 \cdot 1158 \\ &= -204795.744681 - 104184.042553 - 5170.212766 + 314138.297872 \\ &\approx -11.702128 \end{aligned}$$



$$B \approx \begin{bmatrix} 61.914894 \\ 10.425532 \\ 1.170213 \\ -11.702128 \end{bmatrix}$$

$$\text{Marks} \approx 61.914894 + 10.425532 \cdot \text{Hours} + 1.170213 \cdot \text{Sleep} - 11.702128 \cdot \text{GPA}$$

For Student B (Hours=4, Sleep=5, GPA=3.0):

$$\begin{aligned} \text{Predicted} &= 61.914894 + 10.425532 \cdot 4 + 1.170213 \cdot 5 - 11.702128 \cdot 3 \\ &= 61.914894 + 41.702128 + 5.851065 - 35.106384 \approx 74.361702 \end{aligned}$$

$$\text{Marks} = 20 + 5 \cdot (\text{Hours Studied}) + 3 \cdot (\text{Sleep Hours}) - 2 \cdot (\text{Social Media Hours})$$

- 20 = base score (intercept).
- +5 = every 1 extra study hour $\rightarrow$ +5 marks.
- +3 = every extra sleep hour $\rightarrow$ +3 marks.
- -2 = every extra social media hour $\rightarrow$ -2 marks.
- **Do a Quick Numerical Example**
- Student A: 5 hrs studied, 7 hrs sleep, 2 hrs social media
- Student B: 5 hrs studied, 7 hrs sleep, 5 hrs social media

Student A

- Study = 5 hrs $\rightarrow 5 \times 5 = 25$
- Sleep = 7 hrs $\rightarrow 7 \times 3 = 21$
- Social Media = 2 hrs $\rightarrow 2 \times (-2) = -4$
- Base = 20

$$\text{Total} = 20 + 25 + 21 - 4 = 62$$

✓ Student A's score = 62

Student B

- Study = 5 hrs $\rightarrow 25$
- Sleep = 7 hrs $\rightarrow 21$
- Social Media = 5 hrs $\rightarrow 5 \times (-2) = -10$
- Base = 20

$$\text{Total} = 20 + 25 + 21 - 10 = 56$$

✓ Student B's score = 56

Residual/Error

- When you build a regression model, you predict some value $\hat{y}$ (called the **predicted value**).
The **residual** is simply the difference between what really happened and what the model predicted:
- $\text{Residual} = y - \hat{y}$
- If residual = **0**, the prediction was exactly right.
- If residual = **positive**, the model **underestimated** (actual > predicted).
- If residual = **negative**, the model **overestimated** (actual < predicted).

- *“What if we doubled study hours? What happens to marks?”*
- Or: *“What if someone sleeps too much — will the effect always stay positive?”*

Why coefficients can behave/look weird?

- **Multicollinearity (features overlap)**

- Example: “Study hours” and “Chapters read” are highly correlated.
- The model gets *confused about who gets credit*.
- Coefficients may flip signs (positive becomes negative) even if both should help marks.

- **Omitted Variable Bias**

- If you forget an important factor, the model assigns its effect incorrectly.
- Example: Regression of “ice cream sales” vs “drowning deaths” shows positive correlation.
- Missing variable = **temperature** (hot days increase both).

- **Small or Noisy Data**

- With limited data, random variation can distort coefficients.
- Example: If one student who studies a lot still fails (maybe was sick), the coefficient for study hours might appear negative.

Why coefficients can behave/look weird?

- **Interactions Not Captured**

- Linear regression assumes *additive effects only*.
- Example: Sleep improves scores *only if* you've also studied. Without interaction terms, the model might misinterpret.

- **Wrong Functional Form**

- Reality may be nonlinear.
- Example: "Hours of sleep" vs "marks" → too little = bad, moderate = good, too much = bad again (curved relationship).
- A straight-line model forces a misleading positive or negative slope.

- **Scaling or Units**

- If features are measured differently (e.g., minutes vs hours, rupees vs dollars), coefficients may look strange unless standardized.

Linear Regression Model Limitations

- **Linearity Assumption**-It assumes the relationship between predictors (X) and outcome (Y) is **straight-line (linear)**. And real-world data often has **nonlinear patterns**.
- **Sensitivity to Outliers and Variance**-A few extreme values can **pull the regression line** and distort predictions. If variance changes (heteroscedasticity), predictions become unreliable.
- **Multicollinearity**-If predictors (features) are **highly correlated**, coefficients become unstable and hard to interpret.
- **Normality of Errors**-Assumes residuals are normally distributed.

- **Limited to Additive Effects**-Standard linear regression doesn't capture **interactions** or **complex relationships** (e.g., study and sleep may boost marks together more than individually).
- **No Built-in Handling of Missing Data**-Requires **clean, complete data**.
- **Not Always Best for Prediction**-More flexible models (decision trees, random forests, neural networks) may outperform it.
- **Assumes Independent Observations**-Doesn't work well with time series or clustered data unless modifications are made

“Does the straight line still make sense?”

Metrics for Regression

- Metrics measure how close predictions are to actual values.
- **MAE (Mean Absolute Error)** → Average absolute difference.
 - Easy to interpret (“on average, off by X marks”).
- **MSE (Mean Squared Error)** → Penalizes larger errors more.
- **RMSE (Root Mean Squared Error)** → Same units as target, easier to understand.
- **R^2 (Coefficient of Determination)** → How much variance is explained by the model (closer to 1 is better).

R^2 (Coefficient of Determination)

- Explains how much of the variation in the outcome is explained by the model.
- Formula:

$$R^2 = 1 - \frac{\text{Sum of Squared Errors (SSE)}}{\text{Total Sum of Squares (SST)}}$$

- Ranges:
 - $R^2 = 1$: perfect fit
 - $R^2 = 0$: model is no better than just guessing the average
 - Can be **negative** if the model is worse than the mean

Mean Absolute Error (MAE)

Average of absolute errors:

$$MAE = \frac{1}{n} \sum |y - \hat{y}|$$

Easy to explain: "On average, our predictions are off by X marks."

Mean Squared Error (MSE)

Average of squared errors:

$$MSE = \frac{1}{n} \sum (y - \hat{y})^2$$

Penalizes big mistakes more (because of squaring).

Root Mean Squared Error (RMSE)

Square root of MSE:

$$RMSE = \sqrt{MSE}$$

In the **same units** as the outcome (e.g., marks), so easier to interpret.

Activity

$$\text{Marks} \approx 61.914894 + 10.425532 \cdot \text{Hours} + 1.170213 \cdot \text{Sleep} - 11.702128 \cdot \text{GPA}$$

- Actual Marks -> [60,75,85,70,90]
- Predicted Marks -> []
- Calculate Residuals $y - \hat{y}$
- Calculate R^2 , MAE, MSE, RMSE.

SSE (Sum of Squared Errors)

$$\begin{aligned}SSE &= \sum (y - \hat{y})^2 = 0.4255^2 + (-0.4255)^2 + (-0.6383)^2 + (-0.6383)^2 + (0.2766)^2 \\&= 0.1811 + 0.1811 + 0.4075 + 0.4075 + 0.0765 = 1.2537\end{aligned}$$

SST (Total Sum of Squares)

$$SST = \sum (y - \bar{y})^2$$

Mean of actual marks = $\bar{y} = (60 + 75 + 85 + 70 + 90)/5 = 76$.

$$\begin{aligned}SST &= (60 - 76)^2 + (75 - 76)^2 + (85 - 76)^2 + (70 - 76)^2 + (90 - 76)^2 \\&= (-16)^2 + (-1)^2 + (9)^2 + (-6)^2 + (14)^2 = 256 + 1 + 81 + 36 + 196 = 570\end{aligned}$$

$$R^2 = 1 - \frac{SSE}{SST}$$

$$R^2 = 1 - \frac{1.2537}{570} \approx 0.9978$$

$$\begin{aligned} MAE &= \frac{1}{n} \sum |y - \hat{y}| = \frac{0.4255 + 0.4255 + 0.6383 + 0.6383 + 0.2766}{5} \\ &= \frac{2.4042}{5} \approx 0.4808 \end{aligned}$$

$$MSE = \frac{SSE}{n} = \frac{1.2537}{5} = 0.2507$$

$$RMSE = \sqrt{MSE} = \sqrt{0.2507} \approx 0.5007$$